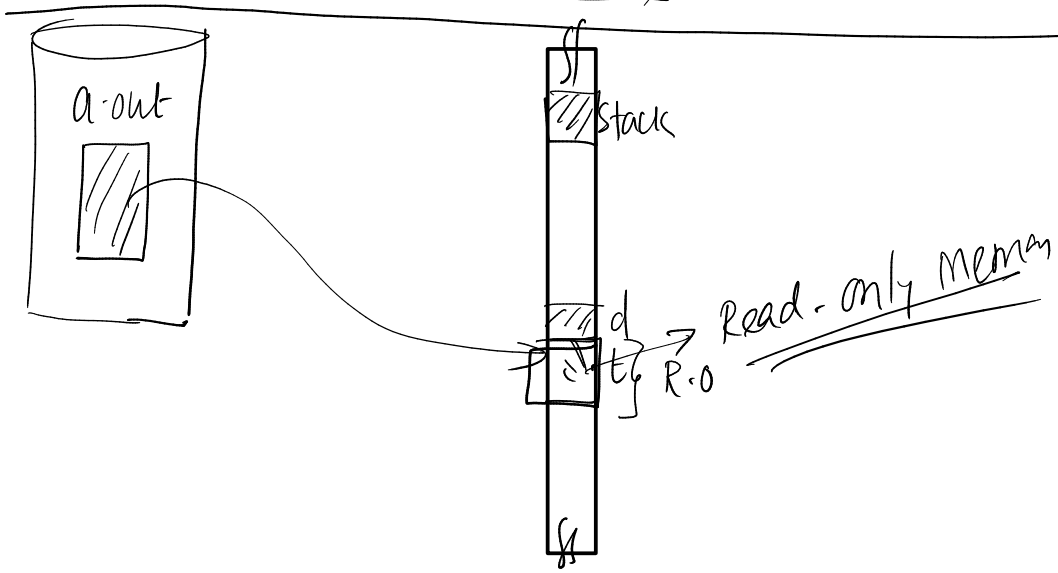
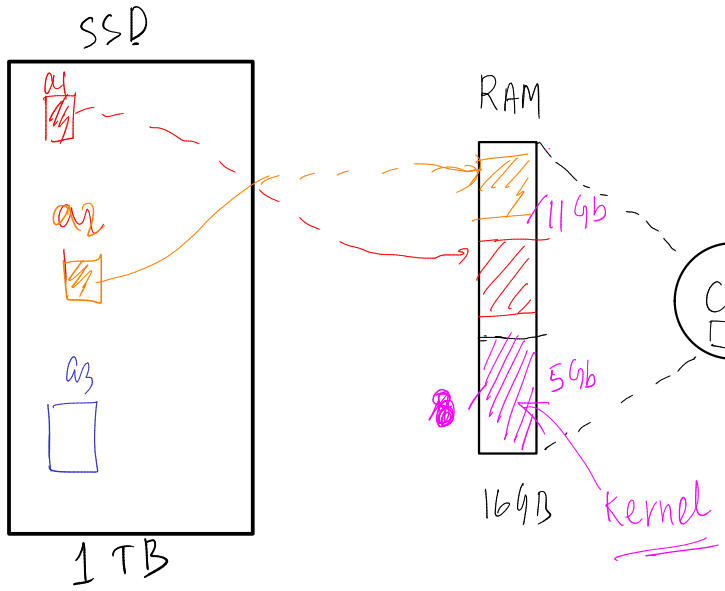
Memory → 1° memory (CPU-register, cache, RAM, Virtual Memory)
→ 2° memory (SSD, HDD, Pendrive, MicroSD, CD, ...).

Q. Is Secondary memory under ROM ?

Does ROM falls under 2° Mem Category
↳ BIOS ROM (EEPROM)





→ 5 Gb of RAM (Approx), ~~used~~ reserved by Kernel.

→ Approximately 11 Gb of RAM available to be used

→ Consider I have 5 apps to run a1 ... a5

a1 → needs 5 Gb

a2 → needs 4 Gb

a3 → needs 6 Gb of mem

Situation: RAM is lesser size than the need of the apps.

OS's mem mgr should manage to execute jobs amounting more than, available RAM.

How? → By Virtual Memory

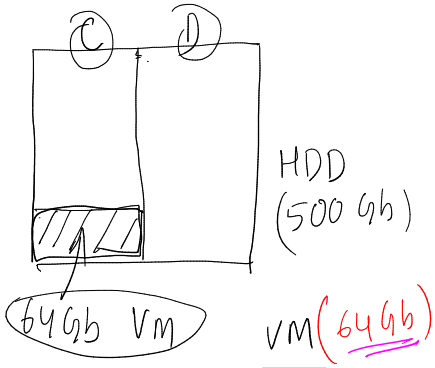
Virtual memory is portion of 2^6 mem
↳ SSD/HDD/pendr

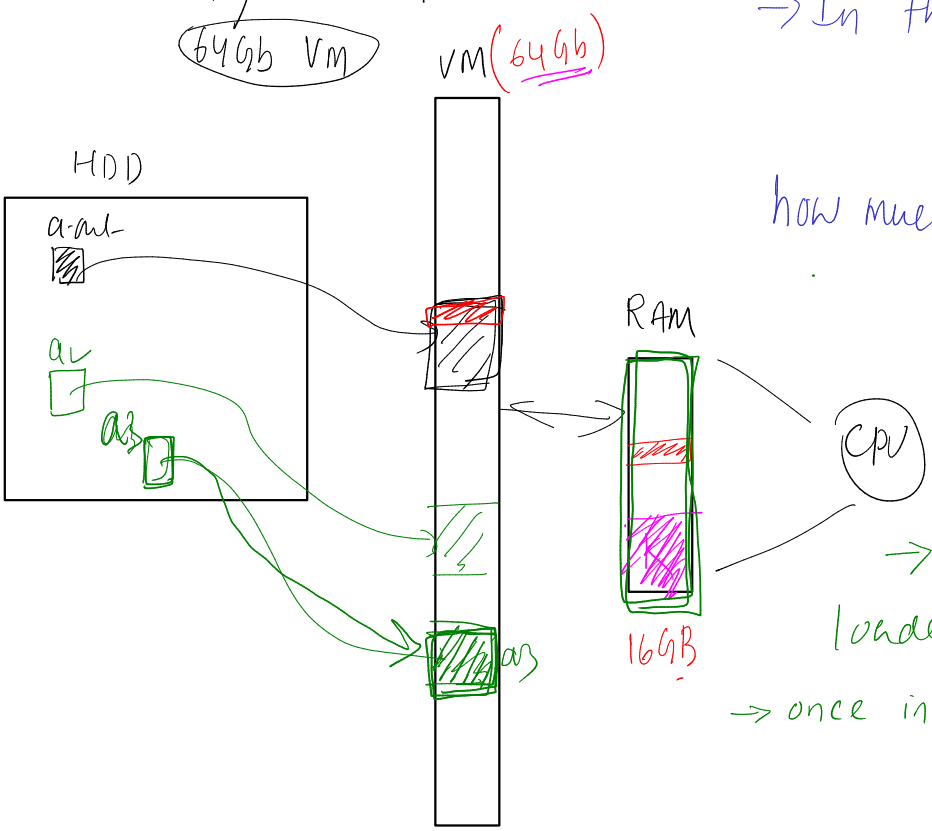
used as if it is RAM.

How is it created?

→ by system Admin Command

→





→ In this scenario (64Gb VM, 16Gb RAM)

how much amount of mem jobs can be executed?

→ once executed, the app is loaded into vm, (virtualized)

→ once in vm, it is a process, has PCB.

Resource Allocator vs Resource Manager

pending → to be discussed

RTOS vs GPOS sched

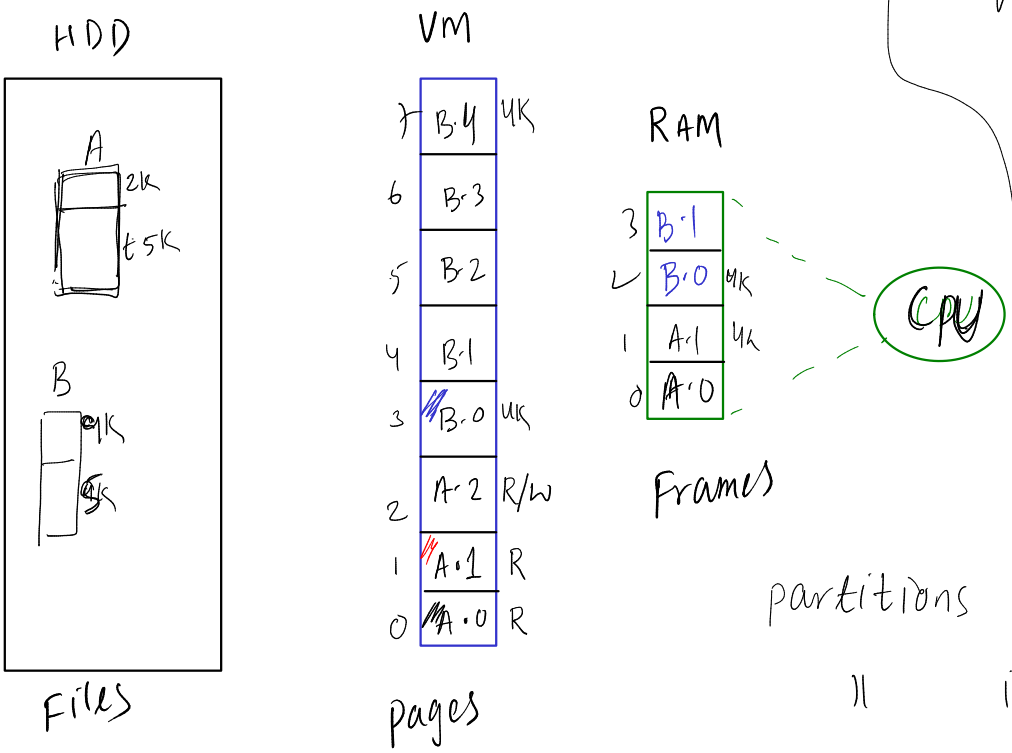
GPOS - Priority - RR.

Linux - Priority sched

OS is a resource manager.

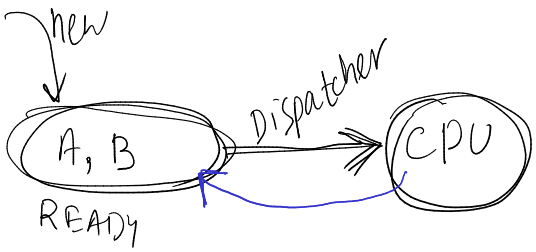
The Paging Method in Mem-mgt

virtual mem may be configured
→ less than RAM
→ Equal to RAM
→ more than RAM



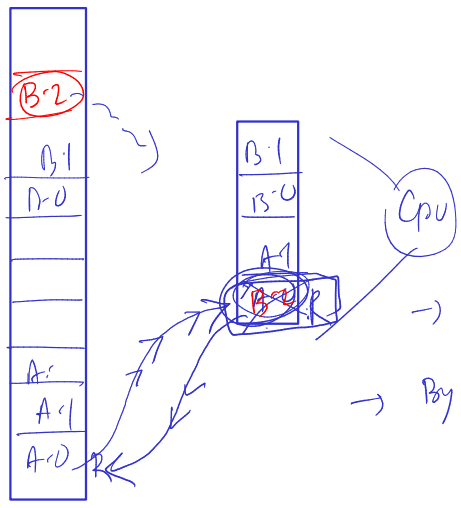
partitions in VM known as Pages

" in RAM (physical mem) known as frames



→ If job A is dispatched for execution - job A must be mapped to RAM.

→ One of the page of A will be loaded into RAM. The page where PC - refers to should - be demanded.



→ B·2 is Demanded.

→ Must be brought to frame.

→ A Page In must occur.

→ A page Replacement will occur.

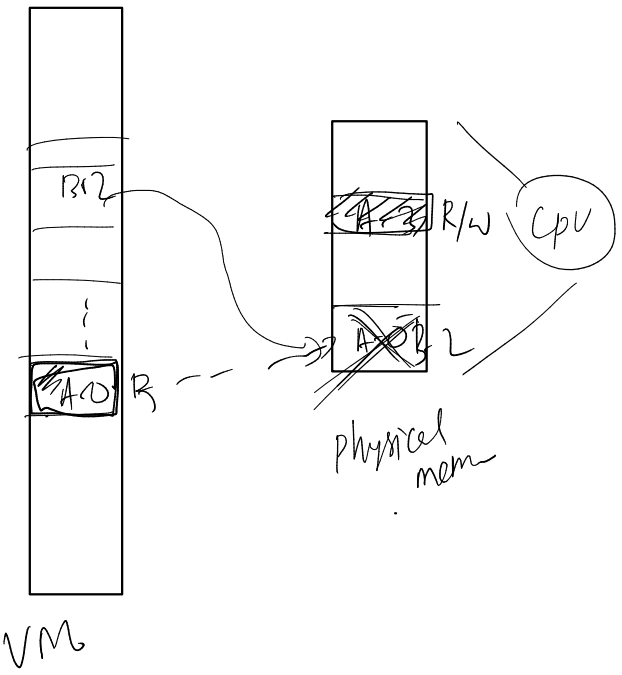
→ By ~~Rep~~ Page-Replacement Algorithm.

→ LRU

→ LU

A·0 is Replaced with B·2, A·0 need not a Page Out

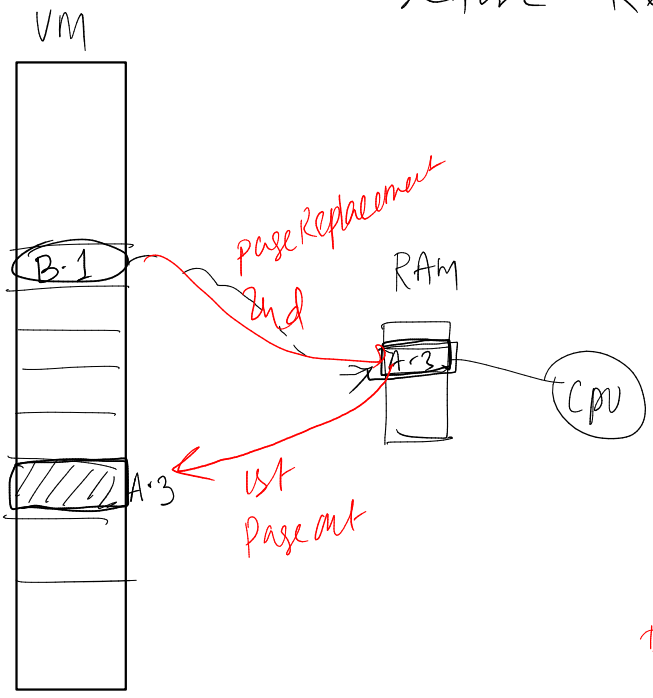
If A R/W Page is replaced, then a Page out required.

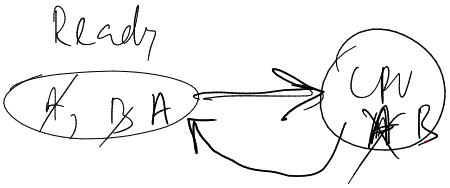


A·0 is getting Replaced overwritten by B·2.

Why R/W page (from the frame) need to be paged out?

→ If R/W Page getting overwritten, then the modified data in that page may be not saved. Hence before Replacement, Page-out will happen





A·3 will be Replaced by B·1